INCH-POUND

MIL-C-47269A (MI)
14 December 1992
SUPERSEDING
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9 August 1974

MILITARY SPECIFICATION

CASTINGS, INSPECTION OF (FOR GUIDED MISSILE AND ROCKET MATERIEL)

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes standards of soundness and freedom from defects for metal castings used in guided missile and rocket materiel and prescribes the requirements for inspection of castings to determine their conformance to such standards.

1.2 Classification. The classification of castings for purposes of inspection hereunder shall be as specified on the design drawing for the castings in accordance with the classifications established in MIL-STD-2175.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1420

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-1949	-	Inspection, Magnetic Particle
MIL-STD-2175	-	Castings, Classification and Inspection of
MIL-STD-6866	-	Inspection, Liquid Penetrant

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM E 155	-	Inspection of Aluminum and Magnesium Castings, Standard Reference Radiographs for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the

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documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Basic requirement. Inspection of castings shall be in accordance with the requirements of MIL-STD-2175 for the applicable class and as required herein.

3.2 Standards. Standards of soundness and freedom from defects which apply for the various areas of a casting shall be as specified on the drawing (see 6.2 and 6.3). These standards shall be used to determine acceptability in accordance with tables I, II and III of this specification. When no standard is specified on the drawing, standard II hereof shall apply. When no standard is specified but the drawing does designate the "critical" areas of a casting, standard II shall apply for the areas so designated, and standard III shall apply for the remainder of the casting.

3.3 Visual inspection.

3.3.1 Gas holes and dross. Maximum acceptable sizes and spacing of visually detected gas holes and dross in the surface of a finished casting shall be as established in table I. Except for Class IIB castings reworked as specified below, defects in excess of those specified in table I shall be cause for rejection.

3.4 Magnetic and penetrant inspection. All castings of ferromagnetic materials shall be inspected in accordance with MIL-STD-1949. All castings of nonferromagnetic materials shall be penetrant inspected in accordance with MIL-STD-6866. Penetrant inspection of aluminum alloy castings which are to be protected by anodic or other chemical film shall be performed before such anodic or chemical process, and shall not be required after the parts are so treated. For castings of all classes, magnetic and penetrant inspection shall be for the purpose of detecting and evaluating cracks and cold shuts only. For Class 1A and Class IIA castings, the disposition of other defects shall be as determined by the results of radiographic inspection. For all classes of castings,

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gas holes and dross exposed on the surface are acceptable if within the limits established in table I.

3.5 Radiographic inspection. Radiographs of casting shall be prepared in accordance with the requirements of MIL-STD-2175 and any applicable radiographic positioning chart. Radiographs of aluminum alloy and magnesium alloy castings, when such are required, shall be compared with the applicable reference radiograph of ASTM E 155 as established in tables II and III respectively (see 6.4). Any defect in excess of those allowed in table II or table III for the particular standard shall be cause for rejection, except that such defect may be accepted in a raw casting if subsequent machining will completely remove the defect.

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TABLE I. Acceptability limits for size and spacing of gas holes and dross on surfaces 1/ of castings.

Applicable standard specified on drawing	Maximum size 2/ of defect for castings classified under MIL-STD-2175			Minimum spacing, edge-to-edge, between defects 3/	Minimum edge distance 3/
	Class 1A	Class IIA	Class IIB		
I	0	0	---	---	---
II	0	0.03125 inch	---	10 D	2.5 D
III	0.03125 inch	0.0625 inch	---	8 D	2 D
none	0.0625 inch	0.0938 inch	0.0938 inch	6 D	1.5 D

NOTES:

1/ Not applicable to surfaces required to be continuous for sealing purposes (e.g. "O" ring groove, etc.); not applicable to threads; not applicable to fillets or corners within the edge distance listed in last column; not applicable to any surface whose width is less than five times the maximum allowable size of defect.

2/ Major diameter or depth of defect, whichever is greater. Notation "O" means no defect is allowed which cannot be removed within drawing dimensional tolerances. Notation --- means no limit is established herein.

3/ "D" is the major diameter of the larger of two neighboring defects when determining spacing of or each single defect when determining edge distance. Notation --- means no limit is established herein.

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TABLE II. Maximum acceptable defects in aluminum alloy casting.

Type of defect	Reference radiograph set in ASTM E 155 Volume I	Maximum allowable defect, by radiograph number in reference set or as otherwise noted		
		Standard I	Standard II	Standard III
Gas holes	1.1	2	4	6
Gas porosity (round)	1.21	1	2	5
Gas porosity (elong.)	1.22	1	4	5
Shrinkage cavity	2.1	1	3	4
Shrinkage porosity or sponge	2.2	1	3	5
Foreign material (less dense material)	3.11	1	3	5
Foreign material (more dense material)	3.12	none	2	2
Segregation	3.2	none	1	1
Hot cracks	4.1	none	none	none
Cold cracks	(4.2)	dwg. tol.	dwg. tol.	dwg. tol.
Cold shut	(4.3)	dwg. tol.	dwg. tol.	dwg. tol.
Surface irregularities	(5.1)	dwg. tol.	dwg. tol.	dwg. tol.
Misruns	(5.2)	dwg. tol.	dwg. tol.	dwg. tol.
Core shift	(5.3)	dwg. tol.	dwg. tol.	dwg. tol.

NOTES:

(1) Notation "none" means no defect of the types illustrated in the reference radiograph set is allowed. Notation "dwg. tol." means that defects of the types illustrated are allowed provided the casting, when machined, conforms to the design drawing and its established tolerances.

(2) Defects in excess of those allowed in this table shall not be cause for rejection if subsequent machining will completely remove them.

(3) Defects allowed by this table shall be cause for rejection when a combination of several defects are determined to be as harmful to mechanical properties as any single rejectable one.

(4) Subsurface gas holes or foreign material allowed by this table shall be cause for rejection when closer than 2 times the maximum dimension to an edge or extremity of the finished part.

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TABLE III. Maximum allowable defects in magnesium alloy casting.

Type of defect	Reference radiograph set in ASTM E 155 Volume II	Maximum allowable defect, by radiograph number in reference set or as otherwise noted		
		Standard I	Standard II	Standard III
Gas holes	1.1	1	2	3
Gas porosity	1.2	none	1	2
Shrink cavity	2.1	none	1	2
Microshrinkage	2.3	1	4	6
Foreign material (less dense material)	3.11	2	4	4
Foreign material (more dense material)	3.12	1	2	2
Hot cracks	4.1	none	none	none
Cold cracks	(4.2)	dwg. tol.	dwg. tol.	dwg. tol.
Cold shut	(4.3)	dwg. tol.	dwg. tol.	dwg. tol.
Surface irregularities	(5.1)	dwg. tol.	dwg. tol.	dwg. tol.
Misrun	(5.2)	dwg. tol.	dwg. tol.	dwg. tol.
Core shift	(5.3)	dwg. tol.	dwg. tol.	dwg. tol.

NOTES:

(1) Notation "none" means no defect of the types illustrated in the reference radiograph set is allowed. Notation "dwg. tol." means that defects of the types illustrated are allowed provided the casting, when machined, conforms to the design drawing and its established tolerances.

(2) Defects in excess of those allowed in this table shall not be cause for rejection if subsequent machining will completely remove them.

(3) Defects allowed by this table shall be cause for rejection when a combination of several defects are determined to be as harmful to mechanical properties as any single rejectable one.

(4) Subsurface gas holes or foreign material allowed by this table shall be cause for rejection when closer than 2 times their maximum dimension to an edge or extremity of the finished part.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

5. PACKAGING

This section not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is intended for use when specifically called out on the design drawing or radiographic positioning chart for the castings involved.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Drawings when required, providing dimensions, tolerances, hardness, and other design details (see 3.2).

6.3 Definitions.

6.3.1 Drawing. The term "drawing" is used throughout this specification to mean the design drawing for the casting involved or the radiographic positioning chart, whichever is applicable.

6.3.2 Standard I. Standard I is primarily intended for highly stressed areas or pressure tightness. Special process control is usually required to meet this standard.

6.3.3 Standard II. Standard II is primarily intended for parts subject to medium stress where better than average quality is required.

6.3.4 Standard III. Standard III is primarily intended for other parts subject to low stressed areas of other castings.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Radiographic inspection
Radiographic standards
Radiography

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-47269A (MI)	921214

3. DOCUMENT TITLE
CASTINGS, INSPECTION OF (FOR GUIDED MISSILE AND ROCKET MATERIEL)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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